Linear Shift (Time)-Invariant Systems

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Systems - Shift Invariance

- **Shift (Time) Invariance:**
  - Shift-Invariant, if for every **input** causing some **output,** every **shifted version by an amount** a **of the input** produces the **same output** but shifted by the amount a.
  - Continuous-time systems:
    - i.e. for system $\mathcal{H}_1$, any input $x(t)$ causes the output $y(t)$,
    - we have for the shifted input $x(t-a)$ causing the output $y(t-a)$.
    - $\boldsymbol{y(t) = \mathcal{H}_1(x(t)) \Rightarrow y(t-a) = \mathcal{H}_1(x(t-a))}$ for all x, t, a.
  - Discrete-time systems:
    - i.e. for system $\mathcal{H}_2$, any input $x[n]$ causes the output $y[n]$,
    - we have for the shifted input $x[n-a]$ causing the output $y[n-a]$.
    - $\boldsymbol{y[n] = \mathcal{H}_2(x[n]) \Rightarrow y[n-a] = \mathcal{H}_2(x[n-a])}$ for all x, n, a.
- Otherwise, we call the system **shift-variant (time-variant)**.

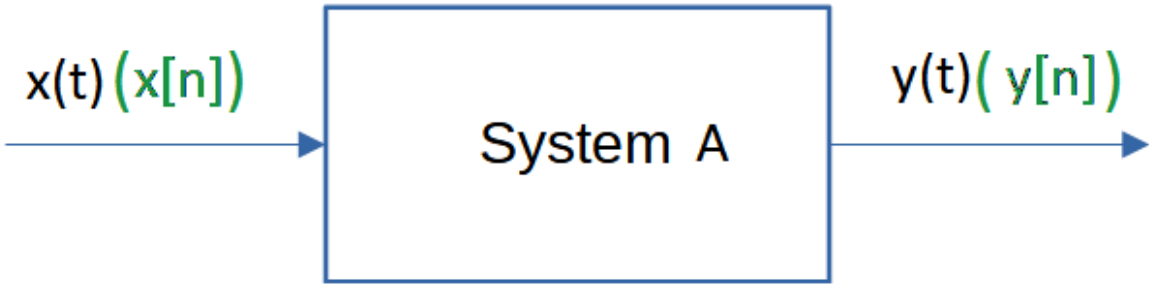
Systems - Linearity

- **Linear Systems:** A system $\mathcal{H}$ is linear if for all input signals $\boldsymbol{x}_1$ and $\boldsymbol{x}_2$ and scalar α, the following hold true:
  - $\mathcal{H}(\boldsymbol{x}_1 + \boldsymbol{x}_2) = \mathcal{H}(\boldsymbol{x}_1) + \mathcal{H}(\boldsymbol{x}_2)$. (Additivity property)
  - $\mathcal{H}(\alpha \boldsymbol{x}_1) = \alpha \mathcal{H}(\boldsymbol{x}_1)$. (Homogeneity property)

- Linearity is same as super-position property.
- Example: $y[n] = 2x[n]$
- What about: $y(t) = x(t) + 1$?

Systems - Response

- To learn about a given **system**,



- we may want to **investigate** how it behaves for a **chosen set of input signals**.
- Which **chosen set**?
- Let's try with **unit impulse signal** - as it is extremely localised?

Systems - Response

- Recall the **sifting property:**

$$
\begin{aligned}
f(t) &= {}_{-\infty}^{\infty} f(u)\delta(u-t)du \\
f[n] &= \sum_{m=-\infty}^{\infty} f[m]\delta[m-n].
\end{aligned}
$$

- **[Obvious]** If we have the samples $f(u)$ and $f[m]$, we can synthesise the original signals $f(t)$ and $f[n]$, by taking **linear combinations of the shifted impulses**.

- But will the **shifted impulses** make the system behave similarly individually?
- What about **Shift-Invariance**?

LSI (LTI) Systems

- Recall the **sifting property:**

$$
\begin{aligned}
f(t) &= {}^{\infty}_{-\infty} f(u)\boldsymbol{\delta(t-u)}du \\
f[n] &= \sum_{m=-\infty}^{\infty} f[m]\boldsymbol{\delta[n-m]}.
\end{aligned}
$$

- **Input signal = Superposition of scaled impulses.**

- **Linear Shift (Time)-Invariant Systems**:
  - **Systems which obey the linearity property and the shift (time)-invariance property.**

LSI (LTI) Systems - Response

- **Continuous-time domain:**
  - For any input $x(t)$ and system $\mathcal{H}$, we have the output $y(t)$ as

$$
\begin{aligned}
y(t) &= \mathcal{H}(x(t)) \\
&= \mathcal{H}(\int_{-\infty}^{\infty} x(u)\delta(t-u)du) \\
&= \int_{-\infty}^{\infty} \mathcal{H}(x(u)\delta(t-u))du, \quad \text{[additivity]} \\
&= \int_{-\infty}^{\infty} x(u) \cdot [\boldsymbol{\mathcal{H}(\delta(t-u))}]du, \quad \text{[homogeneity]} \\
&= \int_{-\infty}^{\infty} x(u) \cdot [\boldsymbol{h_u(t)}]du \\
\ldots & \ldots\ldots\ldots\ldots\ldots\ldots\ldots \\
&= \int_{-\infty}^{\infty} x(u) \cdot \boldsymbol{h(t-u)}du, \quad \text{[shift-invariance]}.
\end{aligned}
$$

LSI (LTI) Systems - Response

- **Discrete-time domain:**
  - For any input $x[n]$ and system $\mathcal{H}$, we have the output $y[n]$ as

$$
\begin{aligned}
y[n] &= \mathcal{H}(x[n]) \\
&= \mathcal{H}\Big(\sum_{m=-\infty}^{\infty} x[m]\delta[n-m]\Big) \\
&= \sum_{m=-\infty}^{\infty} \mathcal{H}(x[m]\delta[n-m]), \quad \text{[additivity]} \\
&= \sum_{m=-\infty}^{\infty} x[m] \cdot [\boldsymbol{\mathcal{H}(\delta[n-m])}], \quad \text{[homogeneity]} \\
&= \sum_{m=-\infty}^{\infty} x[m] \cdot [\boldsymbol{h_m[n]}] \\
\cdots & \cdots\cdots\cdots\cdots\cdots\cdots\cdots \\
&= \sum_{m=-\infty}^{\infty} x[m] \cdot \boldsymbol{h[n-m]}, \quad \text{[shift-invariance]}.
\end{aligned}
$$

Convolution

- **Continuous-time domain:**
  - For any input $x(t)$ and system $\mathcal{H}$, we have the output $y(t)$ as

$$y(t) = {}_{-\infty}^{\infty} x(u) \cdot \boldsymbol{h(t-u)} du.$$

  - If $x(t) = \delta(t)$, we have $y(t) = h(t)$. Hence, the **function $\boldsymbol{h(t)}$** is called its unit impulse response.

- **Discrete-time domain:**
  - For any input $x[n]$ and system $\mathcal{H}$, we have the output $y[n]$ as

$$y[n] = \sum_{m=-\infty}^{\infty} x[m] \cdot \boldsymbol{h[n-m]}.$$

  - If $x[n] = \delta[n]$, we have $y[n] = h[n]$. Hence, the **sequence $\boldsymbol{h[n]}$** is called its unit impulse response.

Convolution

- **Convolution Integral:** [1]

$$y(t) \quad = \quad {}^{\infty}_{-\infty} x(u) \cdot \boldsymbol{h(t-u)} du =: x(t) * h(t) =: (x * h)(t).$$

- **Convolution Sum:**

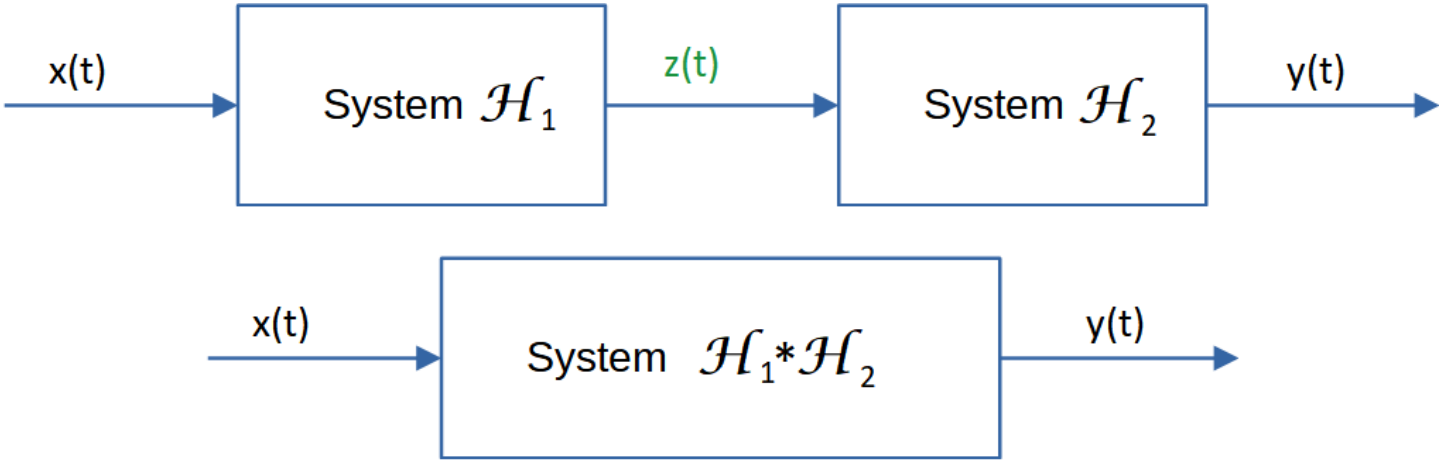
$$y[n] \quad = \quad \sum_{m=-\infty}^{\infty} x[m] \cdot \boldsymbol{h[n-m]} =: x[n] * h[n] =: (x * h)[n].$$

[1] Usual notations, e.g. $(f+g)(t) := f(t) + g(t) = (t) + (t^2) = t + t^2$ where $f(t) := t$ and $g(t) := t^2$ for real t.

Convolution - Properties

- **Associative:**

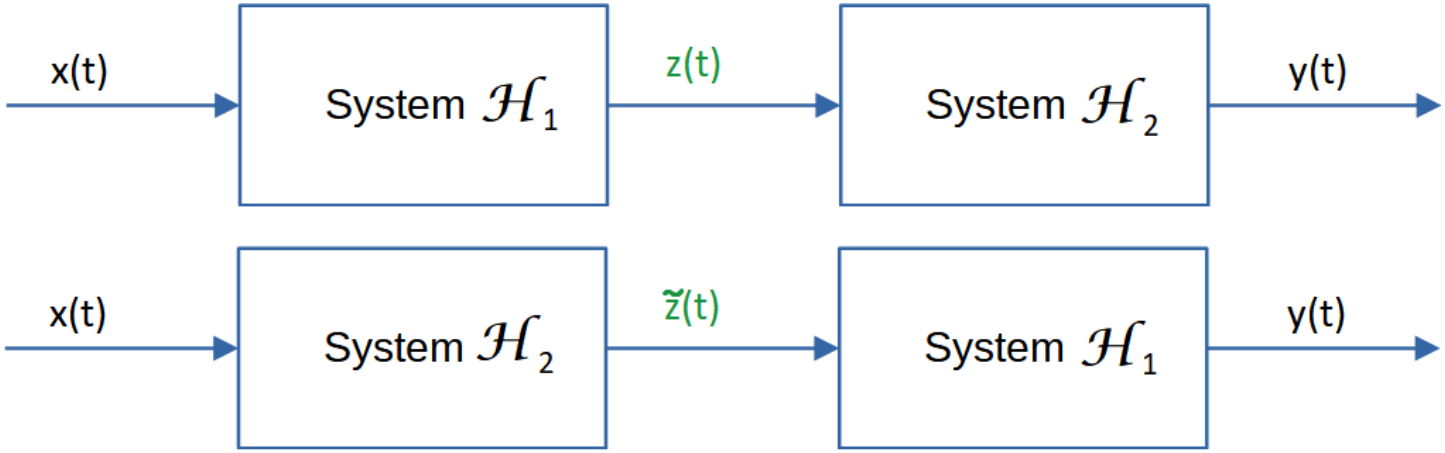
$$
\begin{aligned}
(x(t) * h_1(t)) * h_2(t) &= x(t) * (h_1(t) * h_2(t)) \,. \\
(x[n] * h_1[n]) * h_2[n] &= x[n] * (h_1[n] * h_2[n]) \,.
\end{aligned}
$$



Convolution - Properties

- **Commutative:**

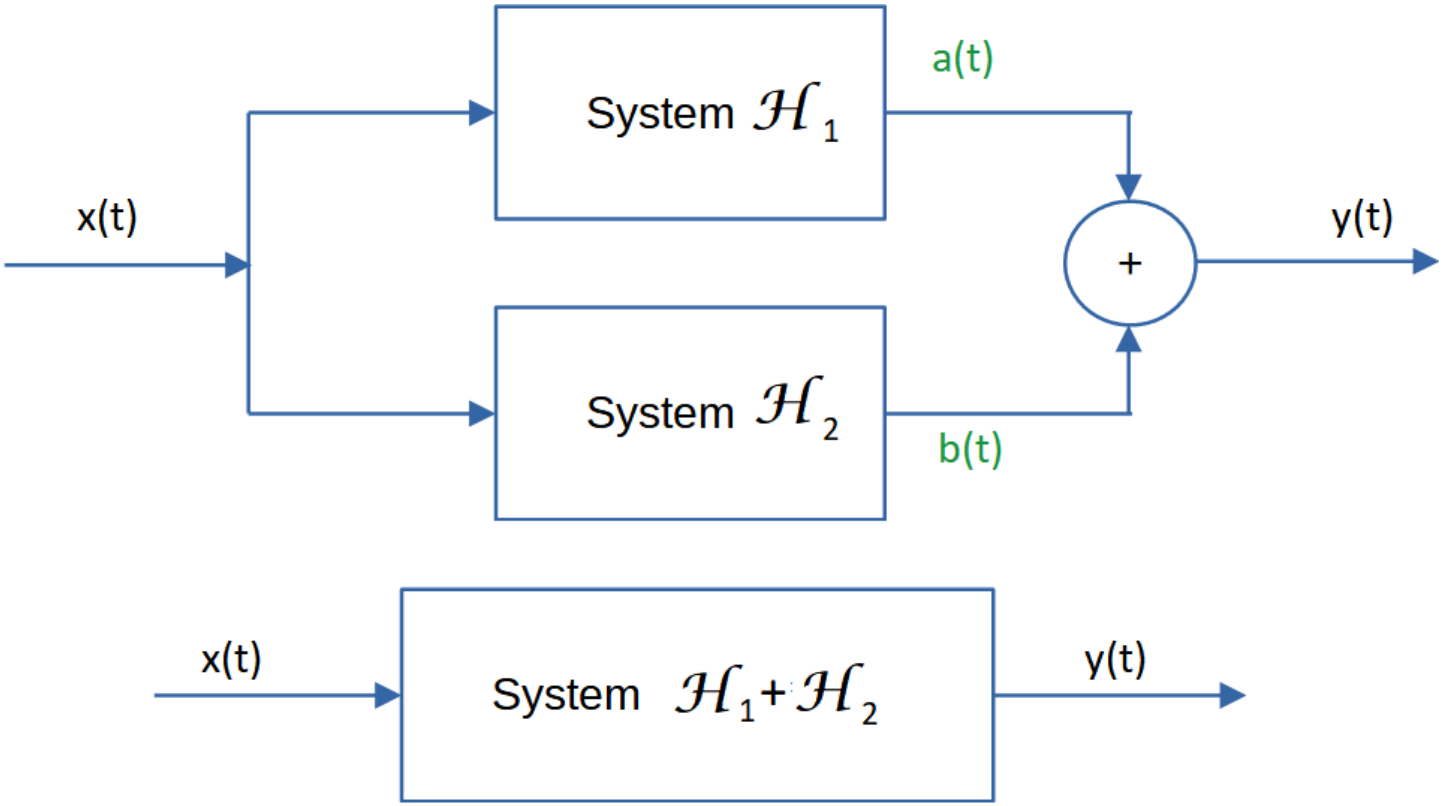
$$
\begin{aligned}
h_1(t) * h_2(t) &= h_2(t) * h_1(t). \\
h_1[n] * h_2[n] &= h_2[n] * h_1[n].
\end{aligned}
$$



Convolution - Properties

- **Distributive:**

$$
\begin{aligned}
x(t) * (h_1(t) + h_2(t)) &= x(t) * h_1(t) + x(t) * h_2(t). \\
x[n] * (h_1[n] + h_2[n]) &= x[n] * h_1[n] + x[n] * h_2[n].
\end{aligned}
$$



LSI (LTI) Systems - Causality

- **Convolution**:

$$y(t) = {}_{-\infty}^{\infty} x(u) \cdot \boldsymbol{h(t-u)} du =: x(t) * h(t).$$

$$y[n] = \sum_{m=-\infty}^{\infty} x[m] \cdot \boldsymbol{h[n-m]} =: x[n] * h[n].$$

- **Causality:**
  - For a system with impulse response h, we have

$$\begin{aligned} h(t) &= 0, \text{ when } t < 0 \\ h[n] &= 0, \text{ when } n < 0. \end{aligned}$$

LSI (LTI) Systems - Stability I

- **For bounded input, output is also bounded.**

- **Continuous-time domain:** (Input bound is $\boldsymbol{B_x}$)

$$
\begin{aligned}
y(t) &= \int_{-\infty}^{\infty} x(u) \cdot \boldsymbol{h(t-u)} du =: x(t) * h(t). \\
|y(t)| &= \left| \int_{-\infty}^{\infty} x(u) \cdot \boldsymbol{h(t-u)} du \right| \\
&\leq \int_{-\infty}^{\infty} |x(u) \cdot \boldsymbol{h(t-u)}| \, du \\
&= \int_{-\infty}^{\infty} |x(u)| \cdot |\boldsymbol{h(t-u)}| \, du \\
&\leq \boldsymbol{B_x} \cdot \int_{-\infty}^{\infty} |\boldsymbol{h(t-u)}| \, du = \boldsymbol{B_x} \cdot \int_{-\infty}^{\infty} |\boldsymbol{h(u)}| \, du.
\end{aligned}
$$

  - So, if impulse response $h(u)$ is **absolutely integrable** ($\|\boldsymbol{h}\|_1 := \int_{-\infty}^{\infty} |\boldsymbol{h(u)}| < \infty$), we have BIBO stability.

LSI (LTI) Systems - Stability I

- **For bounded input, output is also bounded.**

- **Discrete-time domain:** (Input bound is $\boldsymbol{B_x}$)

$$
\begin{aligned}
y[n] &= \sum_{m=-\infty}^{\infty} x[m] \cdot \boldsymbol{h[n-m]} du =: x[n] * h[n]. \\
|y[n]| &= \left| \sum_{m=-\infty}^{\infty} x[m] \cdot \boldsymbol{h[n-m]} \right| \\
&\leq \sum_{m=-\infty}^{\infty} |x[m] \cdot \boldsymbol{h[n-m]}| = \sum_{m=-\infty}^{\infty} |x[m]| \cdot |\boldsymbol{h[n-m]}| \\
&\leq \boldsymbol{B_x} \cdot \sum_{m=-\infty}^{\infty} |\boldsymbol{h[n-m]}| = \boldsymbol{B_x} \cdot \sum_{m=-\infty}^{\infty} |\boldsymbol{h[m]}| .
\end{aligned}
$$

  - So, if impulse response $h[m]$ is **absolutely summable** ($\|\boldsymbol{h}\|_1 := \sum_{m=-\infty}^{\infty} |\boldsymbol{h[m]})| < \infty$), we have BIBO stability.

LSI (LTI) Systems - Stability II

- **For bounded input, output is also bounded.**

- $h(u)$ is **absolutely integrable** and $h[m]$ is **absolutely summable** is sufficient to show BIBO stability.
- Are they necessary?
  - e.g. If we assume $\sum_{m=-\infty}^{\infty} |h[m]| = \infty$ is not finite, can we come up with a bounded input $x[n]$ such that $(x[m] \cdot h[m-n]) = |h[m]|$ for all m, but, at least one value of n?

LSI (LTI) Systems - Memory

- **Convolution**:

$$y(t) = {}_{-\infty}^{\infty} x(u) \cdot \boldsymbol{h(t-u)} du =: x(t) * h(t).$$

$$y[n] = \sum_{m=-\infty}^{\infty} x[m] \cdot \boldsymbol{h[n-m]} =: x[n] * h[n].$$

- **Memoryless:**
  - For an LTI system with impulse response h, we have
    - a)

$$\begin{aligned} h(t) &= 0, \text{ when } t \neq 0 \\ h[n] &= 0, \text{ when } n \neq 0. \end{aligned}$$

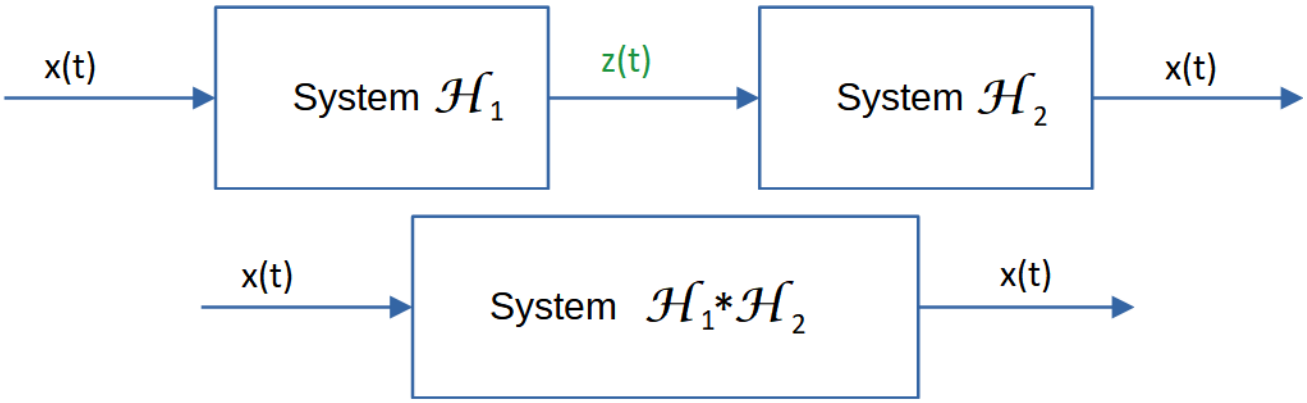
    - b)

$$\begin{aligned} h(0) &= K \\ h[0] &= K' \end{aligned}$$

and for some constants K and K'.

LSI (LTI) Systems - Invertibility

- **Forward System $\mathcal{H}_1$**
  - Represented by convolution.
- **How to find invertibility?**



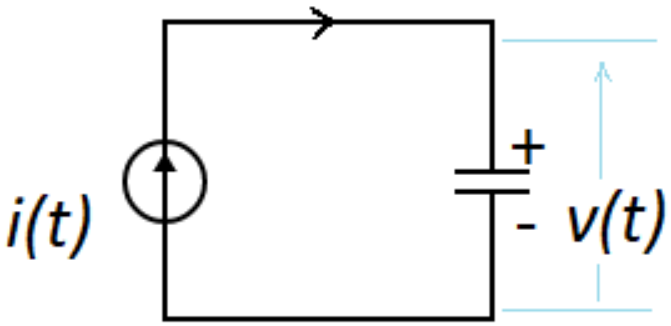
- **How about $\mathcal{H}_1 * \mathcal{H}_2 = \mathcal{I}d$?**
  - $h_1(t) * h_2(t) = \delta(t)$
  - $h_1[n] * h_2[n] = \delta[n]$

Linear Constant Co-efficient Differential Equations

- Representations of numerous systems.
- The relationships between **inputs and outputs** are mentioned **implicitly**.
- To solve them, need more information like initial conditions, boundary conditions.
- **Homogeneous Solution+Particular Solution.**
- Similarly for **Difference Equations.**

Linear Constant Co-efficient Differential Equations

- Electrical Capacitor circuit



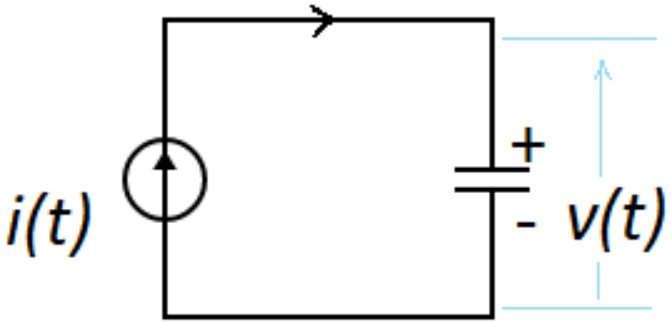
- $v(t) = {}^{t}_{-\infty} Ci(t') \cdot dt' =$

$$\underbrace{{}^{0}_{-\infty} Ci(t') \cdot dt'}_{\text{output-state just before t=0}} + \underbrace{{}^{t}_{0} Ci(t') \cdot dt'}_{\text{response from t=0}}$$

- What is **impulse response** $h(t)$?
- Generally, it is assumed at some time assigned $t = 0$, the input signal $i(t)$ is applied (meaning $i(t) = 0, t < 0$).

Linear Constant Co-efficient Differential Equations

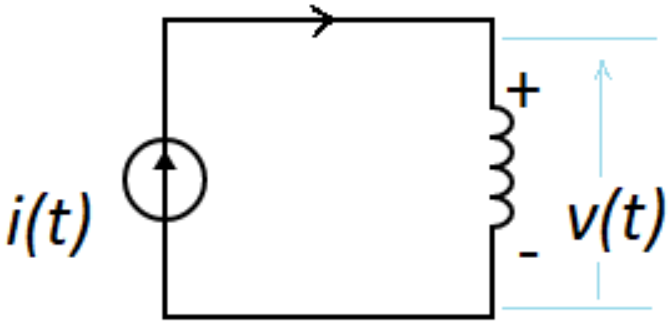
- Electrical Capacitor circuit



- Relationship: $v(t) = {}^{t}_{-\infty} Ci(t') \cdot dt'$
- What is **step response** $v_{step}(t)$ for input $i(t) = Iu(t)$?
- We then have
  $v_{step}(t) = \underbrace{{}^{t}_{0} CIu(t') \cdot dt'}_{\text{response from t=0}} = CI\,{}^{t}_{0}\, u(t') \cdot dt' = CIt = CIr(t)$ for $t \geq 0$.
- **Ramp signal** $\boldsymbol{r(t) := \begin{cases} 0, & t < 0 \\ t, & t \geq 0 \end{cases}}$.

Linear Constant Co-efficient Differential Equations

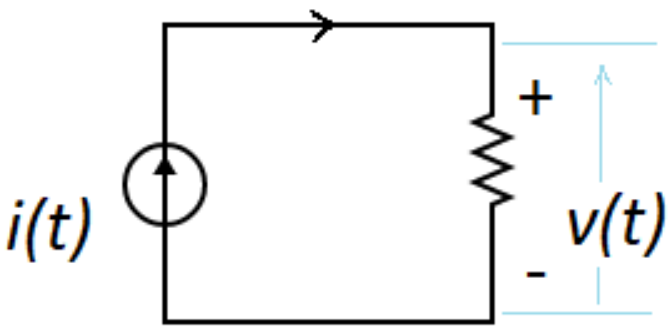
- Electrical Inductor circuit



- Relationship: $v(t) = L\frac{d}{dt}i(t)$
- What is **step response** $v_{step}(t)$ for input $i(t) = Iu(t)$?
- We then have $v_{step}(t) = L\frac{d}{dt}(Iu(t)) = LI\frac{d}{dt}u(t) = LI\delta(t)$.
- What is impulse response $h(t)$?
- **Unit doublet signal**: $\boldsymbol{u_1(t) := \frac{d}{dt}\delta(t)}$.

Linear Constant Co-efficient Differential Equations

- Electrical Resistor circuit



- Relationship: $v(t) = Ri(t)$
- What is **step response** $v_{step}(t)$ for input $i(t) = Iu(t)$?
- We then have $v_{step}(t) = RIu(t)$.
- What is **impulse response** $h(t)$?

Linear Constant Co-efficient Differential Equations

- **In general, we have** relationship:

$$\sum_{k=0}^{N} a_k \frac{d^k}{dt^k} y(t) = \sum_{k=0}^{M} b_k \frac{d^k}{dt^k} x(t)$$

- **Initial conditions**: $y(t_0), y'(t_0), \cdots, y^{(N-1)}(t_0)$.
- Order is N. Solution has two parts:
  - One due to homogeneous $\sum_{k=0}^{N} a_k \frac{d^k}{dt^k} y(t) = 0$. **(Natural response)**
  - Other explicitly due to input $x(t)$. **(Forced response / Particular solution)**
- **Conditions for LTI? Causal?**

Linear Constant Co-efficient Difference Equations

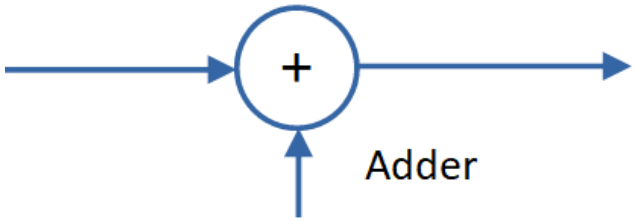
- **In general, we have** relationship:

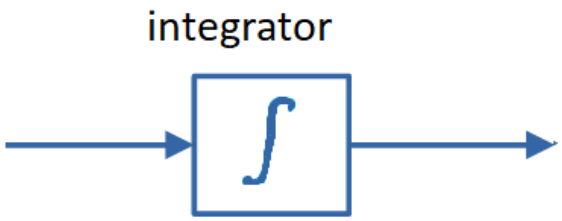
$$\sum_{k=0}^{N} a_k y[n-k] = \sum_{k=0}^{M} b_k x[n-k]$$

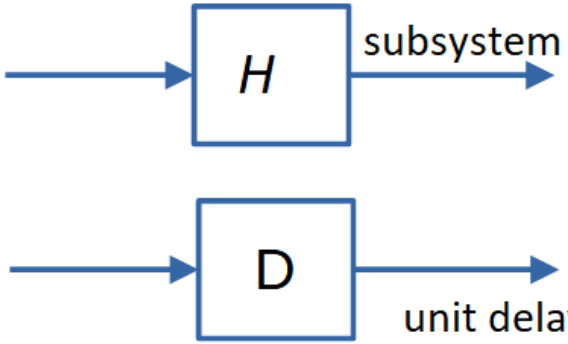
- **Initial conditions**: $y[n_0], y[n_0-1], \cdots, y[n_0-N+1]$.
- Order is N. Solution has two parts:
  - One due to homogeneous $\sum_{k=0}^{N} a_k y[n-k] = 0$. **(Natural response)**
  - Other explicitly due to input $x[n]$. **(Forced response / Particular solution)**
- **Conditions for LTI? Causal?**

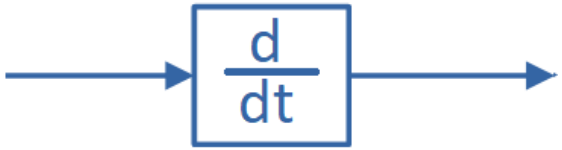
Block Diagrams

- **Block diagrams** - pictorial representations of linear constant co-efficient differential and difference equations with blocks and connections.





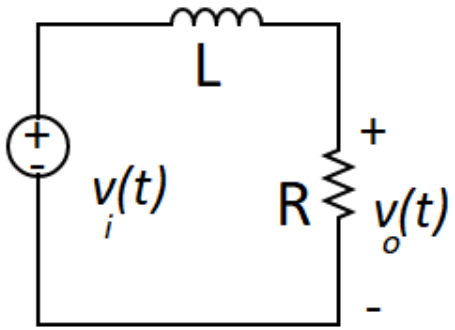




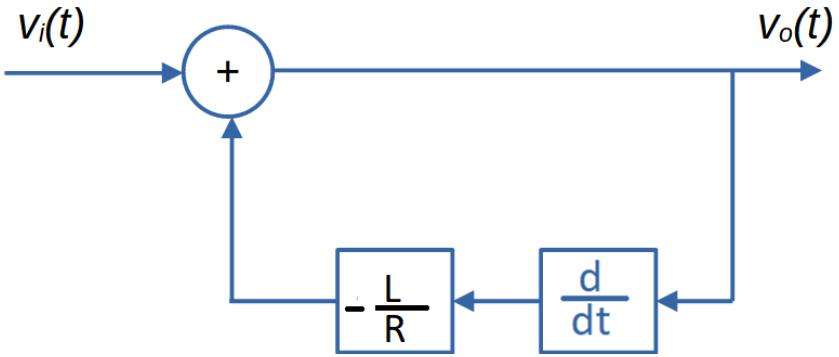
differentiator

Block Diagrams

- Example: LR series circuit with input $v_i(t)$ and output $v_o(t)$ as



- $v_o(t) + \frac{L}{R}\frac{d}{dt}v_0(t) = v_i(t)$ can be re-written as
  $v_o(t) = v_i(t) - \frac{L}{R}\frac{d}{dt}v_0(t)$



Signal Flow Diagrams

- **Signal Flow diagrams** - another pictorial representations of linear constant co-efficient differential and difference equations with signal nodes, junctions and multipliers.